

Ziquan Wei

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EDUCATION

- University of North Carolina at Chapel Hill, USA *PhD in Computer Science, Aug 2023 - May 2028*
- Huazhong University of Science and Technology, China *Master in Biomedical Engineering, Sept 2019 - June 2022*
- Yunnan Normal University, China *Bachelor in Computer Science, Sept 2015 - June 2019*

RESEARCH INTEREST

Digital twins of the human body: generative models of structure that deforms over time, built for one individual from routinely measured signals and able to be run forward under altered conditions. I work on 4D generation of deforming anatomy from sparse physiological conditioning, on non-rigid correspondence between deforming shapes, and on latent spaces whose samples stay on a matrix manifold by construction rather than by projection.

RESEARCH EXPERIENCE

- University of North Carolina at Chapel Hill  *Research Assistant, Aug 2023 - Current*
 - **Conditional 4D Motion Generation:** Built a mechanics foundation model that generates individualized 4D cardiac motion, one deforming mesh sequence per subject across the full cycle, conditioned only on the surface ECG, recovering dense moving geometry from a sparse boundary measurement [5].
 - **Manifold-Constrained Generative Models:** Designed SPD-VQVAE, a vector-quantized autoencoder over a Cholesky factorization that keeps every generated matrix inside the symmetric positive-definite cone with no projection or repair step, with a proof it avoids $O(N^3)$ cost (Theorem 3.1); used inside a conditional latent diffusion model over tokenized event sequences linked to multi-organ imaging on UK Biobank (~500K subjects, 1,000+ diseases) that supports scenario generation under an altered mediating state [3]. Second author on conditional diffusion with ordinal regression for longitudinal generation [16].
 - **Geometric Deep Learning at Scale:** Graph transformers that model pathway structure rather than pairwise edges [18][4]; a Riemannian manifold mixer built on the scattering transform [20]; 1.2B-parameter pretraining [1] and a Mixture-of-Experts encoder robust across acquisition sites and scanners [14].
- EGRA.ai  *Research Intern, May 2026 - Aug 2026*
 - **RL post-training:** Trained a reward model on physiological feedback signals to supplement human preference labels, and ran GRPO policy optimization against a programmatically verifiable composite reward.
- University of North Carolina at Chapel Hill *Predoctoral Fellow, Aug 2022 - Aug 2023*
 - **Teravoxel Volumetric Pipelines:** Built 3D instance segmentation at teravoxel scale, including recovery of context lost at tile boundaries, shipped as the CellPheno platform over a 118-volume whole-brain lightsheet cohort [24][6].
- Huazhong University of Science and Technology *Graduate Researcher (MSc), Sept 2019 - June 2022*
 - **High-Throughput Gigapixel Inference:** First-authored a detection framework that processes a 4-gigapixel whole slide image in 55.9s by cutting redundant forward passes by two orders of magnitude [27], and a detector that holds up on out-of-distribution slides [26].
- Nanyang Technological University, Singapore *Research Intern, Jun 2021 - Sept 2021*
 - **Audio-Visual Human Behavior:** Co-first-authored a leader-follower attentive fusion of audio and visual streams for continuous emotion recognition [28].
- Yunnan Normal University *Research Assistant, Sept 2016 - June 2019*
 - **Non-Rigid Registration and Deformable Correspondence:** First-authored multi-temporal registration of UAV imagery over deforming terrain [32]; co-authored non-rigid point set registration with a dual-feature finite mixture model and global-local structural preservation [31], and non-rigid image registration with dynamic Gaussian component density and space curvature preservation [30].

SKILLS

- **Geometry & 4D:** 4D mesh sequence generation from sparse conditioning, non-rigid and point set registration, deformable correspondence, 3D instance segmentation in volumetric data, Riemannian and symmetric positive-definite manifold constraints, graph transformers, scattering transforms
- **Generative & Large-Scale Modeling:** Conditional and latent diffusion, vector-quantized autoencoders, Mixture-of-Experts, 1.2B-parameter pretraining, reward modeling and GRPO, LLM agents with MCP tool use, multi-GPU distributed training
- **Engineering:** Python, PyTorch, C++, TypeScript / React, FastAPI, Nextflow, Ollama, multi-GPU distributed training, HPC administration (10 nodes, 20 GPUs, 1.5PB)

- [1] Wei Z., Dan T., Wu G. **Large Connectome Model: An fMRI Foundation Model of Brain Connectomes Empowered by Brain-Environment Interaction in Multitask Learning Landscape.** *AAAI* '26.
- [2] Wei Z., Wu G. **Teravoxel Microscopy Image Analysis for Neurological Diseases.** *ARBME* '26 (IF=11.5).
- [3] Wei Z., Dan T., Wu G. **Marrying Generative Model of Healthcare Events with Digital Twin of Social Determinants of Health for Disease Reasoning.** *ICML* '26.
- [4] Wei Z., Dan T., Ding J., Laurienti P.J., Wu G. **NeuroDetour: A Neural Pathway Transformer for Uncovering Structural-Functional Coupling Mechanisms in Human Connectome.** *MedIA* '26 (IF=14.0).
- [5] Wei Z., et al. **From Heartbeat to Cardiochoreography: A Mechanics Foundation Model for Individualized 4D Cardiac Motion Generation Conditioned on Electrophysiology.** *NeurIPS* '26 (In submission, ratings 5 4 4).
- [6] Wei Z., Curtin I., Kyere F.A., et al. **CellPheno: A High-throughput Computational Platform for Quantifying Cellular Resolution Whole Brain Microscopy Images.** *bioRxiv* '26.
- [7] Ding J., Dan T., Wei Z., Laurienti P.J., Wu G. **Scanning the Horizon of Replicability in Neuroscience: A Recipe of Developing Replicable Deep Models for Functional Neuroimages.** *IEEE TBME* '26 (IF=4.5).
- [8] Wen T., Wei Z., Dan T., et al. **Image Imputation for Real-World Defective Microscopy Images of Mouse Brains.** *ISBI* '26.
- [9] Ding J., Dan T., Wei Z., Laurienti P.J., Wu G. **Machine learning on dynamic functional connectivity: Promise, pitfalls, and interpretations.** *Inf. Sci.* '26 (IF=6.0).
- [10] Chen Y., Li Y., Tong J., ..., Wei Z., et al. **NeuroPilot: An Agent-Driven Smart Pipeline for Processing, Quality Control, and Managing Neuroimages.** *arXiv* '26.
- [11] Kyere F.†, Curtin I.†, Wei Z.†, et al. **Chd8 haploinsufficiency leads to molecular layer heterotopias and age-dependent cortical expansion.** *bioRxiv* '26.
- [12] Wei Z., Dan T., Ding J., Wu G. **Efficient Graph Representation Learning by Non-Local Information Exchange.** *Electronics* '25 (IF=2.9).
- [13] Wei Z., Dan T., Wu G. **Brain-Environment Cross-Attention (BECA) Meta-matching: A New Perspective of Brain Connectome Zero-Shot Learning.** *MICCAI* '25 (Early accept (9%)).
- [14] Wei Z., Dan T., Chen T., Wu G. **BrainMoE: Cognition Joint Embedding via Mixture-of-Expert Towards Robust Brain Foundation Model.** *NeurIPS* '25.
- [15] Cho H., Wei Z., Lee S., et al. **Conditional Diffusion Model using Ordinal Regression for Longitudinal Neurodegenerative Data Generation.** *Alz. & Dem.* '25 (IF=12.8).
- [16] Cho H., Wei Z., Lee S., et al. **Conditional Diffusion with Ordinal Regression: Longitudinal Data Generation for Neurodegenerative Disease Studies.** *ICLR* '25 (Spotlight poster (3.26%)).
- [17] Wei Z., Dan T., Ding J., Laurienti P., Wu G. **Representing Functional Connectivity with Structural Detour: A New Perspective to Decipher Structure-Function Coupling Mechanism.** *MICCAI* '24.
- [18] Wei Z., Dan T., Ding J., Wu G. **NeuroPath: A Neural Pathway Transformer for Joining the Dots of Human Connectomes.** *NeurIPS* '24.
- [19] Wei Z., Wu G. **Non-local Exchange: Introduce Non-locality via Graph Re-wiring to Graph Neural Networks.** *NeurIPS-W* '24.
- [20] Dan T., Wei Z., Kim W.H., Wu G. **Exploring the Enigma of Neural Dynamics Through A Scattering-Transform Mixer Landscape for Riemannian Manifold.** *ICML* '24.
- [21] Ding J., Dan T., Wei Z., Laurienti P., Wu G. **A Wasserstein Recipe for Replicable Machine Learning on Functional Neuroimages.** *MICCAI* '24.
- [22] Wu Y., Liu Y., Li Y., ..., Wei Z., et al. **Symmetry engineering in 2D bioelectronics facilitating augmented biosensing interfaces.** *PNAS* '24 (IF=9.5).
- [23] Wei Z., Dan T., Ding J., et al. **High Throughput Deep Model of 3D Nucleus Instance Segmentation by Stereo Stitching Contextual Gaps.** *ISBI* '23.
- [24] Wei Z., Dan T., Ding J., Dere M., Wu G. **A General Stitching Solution for Whole-Brain 3D Nuclei Instance Segmentation from Microscopy Images.** *MICCAI* '23 (Early accept (14%)).
- [25] Dan T., Ding J., Wei Z., et al. **Re-think and re-design graph neural networks in spaces of continuous graph diffusion functionals.** *NeurIPS* '23.
- [26] Wei Z., Cheng S., Cai J., et al. **Cervical Glandular Cell Detection from Whole Slide Image with Out-Of-Distribution Data.** *arXiv* '22.
- [27] Wei Z., Cheng S., Hu J., et al. **An Efficient Cervical Whole Slide Image Analysis Framework Based on Multi-scale Semantic and Location Deep Features.** *arXiv* '21.
- [28] Zhang S.†, Ding Y.†, Wei Z.†, Guan C. **Continuous emotion recognition with audio-visual leader-follower attentive fusion.** *ICCVW* '21.
- [29] Cheng S., Liu S., Yu J., ..., Wei Z., et al. **Robust whole slide image analysis for cervical cancer screening using deep learning.** *Nat. Comm.* '21 (IF=18.1).
- [30] Yang Z., Yang Y., Yang K., Wei Z.. **Non-rigid image registration with dynamic Gaussian component density and space curvature preservation.** *IEEE TIP* '18 (IF=13.7).
- [31] Zhang S., Yang K., Yang Y., Luo Y., Wei Z.. **Non-rigid point set registration using dual-feature finite mixture model and global-local structural preservation.** *Pattern Recognit.* '18 (IF=9.1).
- [32] Wei Z., Han Y., Li M., et al. **A small UAV based multi-temporal image registration for dynamic agricultural terrace monitoring.** *Remote Sens.* '17 (IF=4.3).

HONORS & SERVICE

- **Honors:** Editor of Distinction Award 2025, Scientific Reports (editorial board); Graduate School Transportation Grant, UNC-Chapel Hill; Outstanding Graduate, Huazhong University of Science and Technology
- **Conference Reviewer:** NeurIPS, ICML, ICLR, AAAI, AISTATS, MICCAI
- **Journal Reviewer:** IEEE Transactions on Medical Imaging, NeuroImage, Pattern Recognition

TALKS

- 2025 Tissue Clearing & Imaging Conference, UNC-Chapel Hill  Speaker, Apr 2025
- **CellPheno:** A High-throughput Computational Platform of Cellular Phenotyping for Real-world Whole Brain Microscopy Images.